Functional and differential equations – 2

Functional-differential equations

13 сентября 2026 года

Theoretical minimum.

Fact 1 (three shapes). A functional-differential equation relates the values of f and of its derivative at *different* points. Three shapes cover almost everything: a shift, $f'(x) = F\big(f(x+a)\big)$ or $f(x) = \int_{x-1}^{x} f$; a composition, $f'(x) = f(f(x))$; and a mean value, $f(a) - f(b) = (a-b)f'(\mu(a,b))$ for a prescribed mean μ.

Theorem 1 (the mean value equations). If f is differentiable and $f(a) - f(b) = (a-b)f'\big(\frac{a+b}{2}\big)$ for all a, b, then f is a polynomial of degree at most two; for the geometric mean the answer is different and is found by first showing that f' is determined on a scaling orbit. In both cases the method is the same: fix one variable, differentiate in the other, and read the resulting ordinary differential equation.

Theorem 2 (shifts raise regularity). If f is continuous and $f(x) = \int_{x-1}^{x} f$, then f is C^∞, and $f'(x) = f(x) - f(x-1)$; every relation of this type is differentiated once and becomes a delay equation, whose solutions are found by substituting $e^{\lambda x}$ and analysing the characteristic equation — which now has infinitely many roots.

Fact 2 (compositions are rigid). An equation such as $f'(x) = f(f(x))$ admits almost no solutions: monotonicity and sign considerations usually force a contradiction. The standard argument compares the growth of the two sides and uses that a composition inherits monotonicity, as in problem 7 of the homework of Differentiable functions – 1.

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1. (Карасёв 4.5) Find all differentiable functions $f\colon \mathbb{R} \to \mathbb{R}$ satisfying $f(a) - f(b) = (a-b)f'\big(\frac{a+b}{2}\big)$ for all $a, b \in \mathbb{R}$.

2. (Карасёв 4.6) Find all differentiable functions $f\colon (0,+\infty) \to \mathbb{R}$ satisfying $f(a) - f(b) = (a-b)f'\left(\sqrt{ab}\right)$ for all positive a and b.

3. (ISTCiM 2024) Find all differentiable functions $f\colon \mathbb{R} \to \mathbb{R}$ satisfying $f(x+y) = f'(x)f(y) + f(x)f'(y)$ for all $x, y \in \mathbb{R}$.

4. (RMO) Find all functions $f\colon \mathbb{R} \to \mathbb{R}$ such that f^2 is differentiable and $f = (f^2)'$.

5. Let f be continuous on $[0,\infty)$ with $f(x) = \int_{x-1}^{x} f(t)\,dt$ for all $x \geqslant 1$. Prove that f has k continuous derivatives on (k,∞) for every k, find the characteristic equation of the corresponding delay equation, and describe the bounded solutions. (That such an f has bounded variation is the subject of problem 5 in the homework of Definite integrals – 4.)